Experiment 1(b): Enabling Multi-Factor Authentication (MFA) to Secure Access

Aim

To create an IAM user in AWS, sign in as that user, and enable Multi-Factor Authentication (MFA) so that an extra verification code is needed every time the user logs in.

Part A: Creating an IAM user

Step 1: Sign in as the root user

I opened the browser, went to the AWS Management Console and signed in with my root user email and password.

Step 2: Open IAM

In the search bar I typed IAM and opened Identity and Access Management (IAM). From the left menu I clicked Users (IAM users). The list was empty, so I clicked Create user.

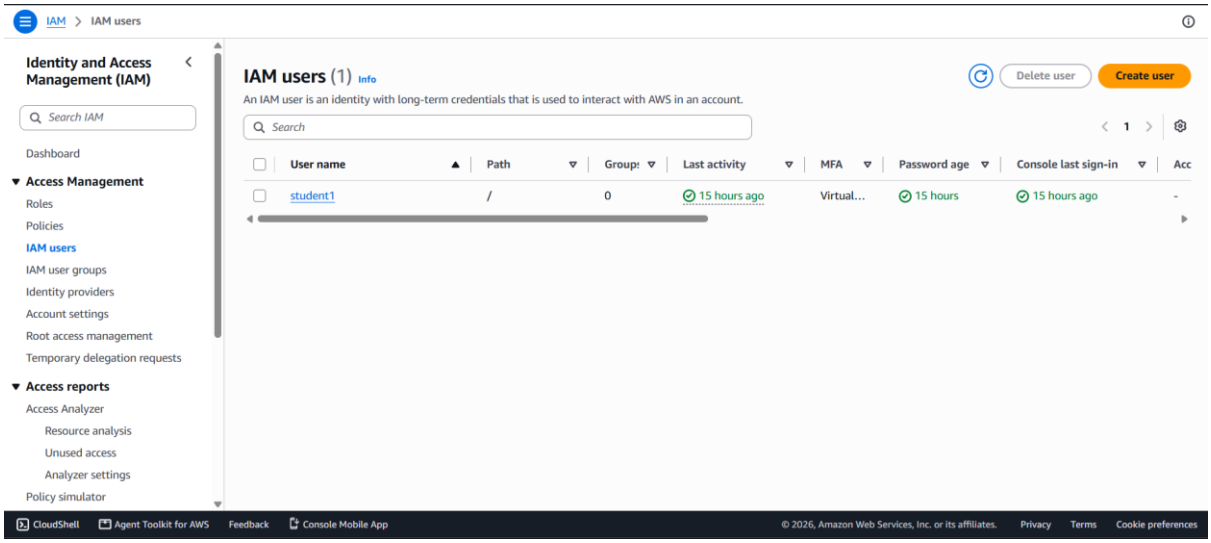
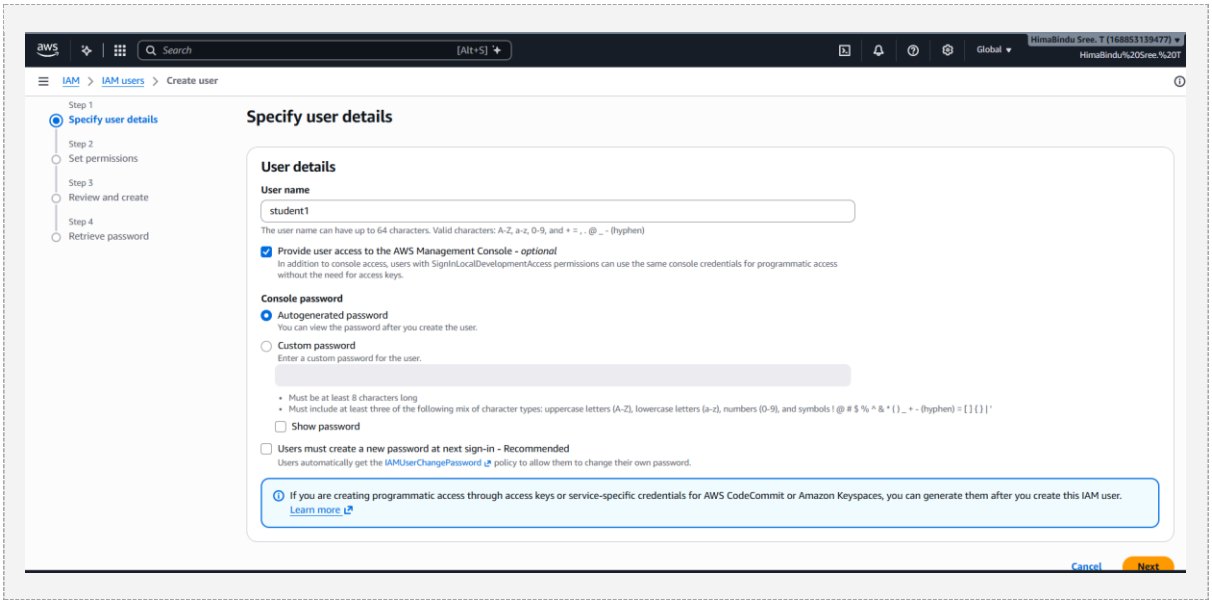


Figure 1: IAM users page with the Create user button

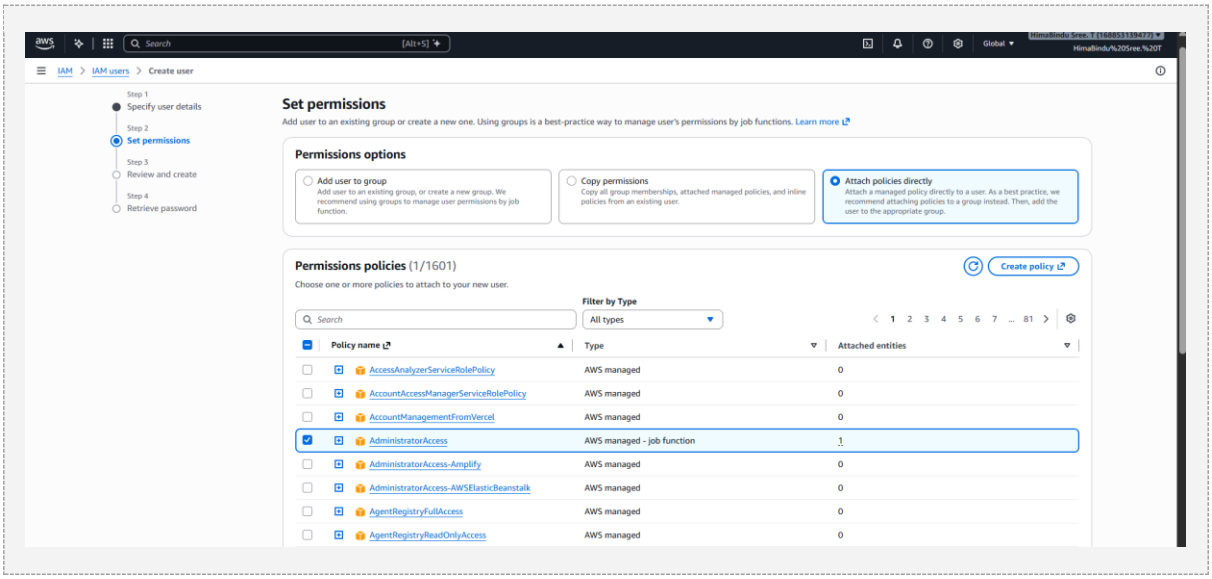
Step 3: Enter the user details

I typed a username (student1) and ticked Provide user access to the AWS Management Console. I chose Autogenerated password and ticked the option that the user must create a new password at the next sign-in, then clicked Next.



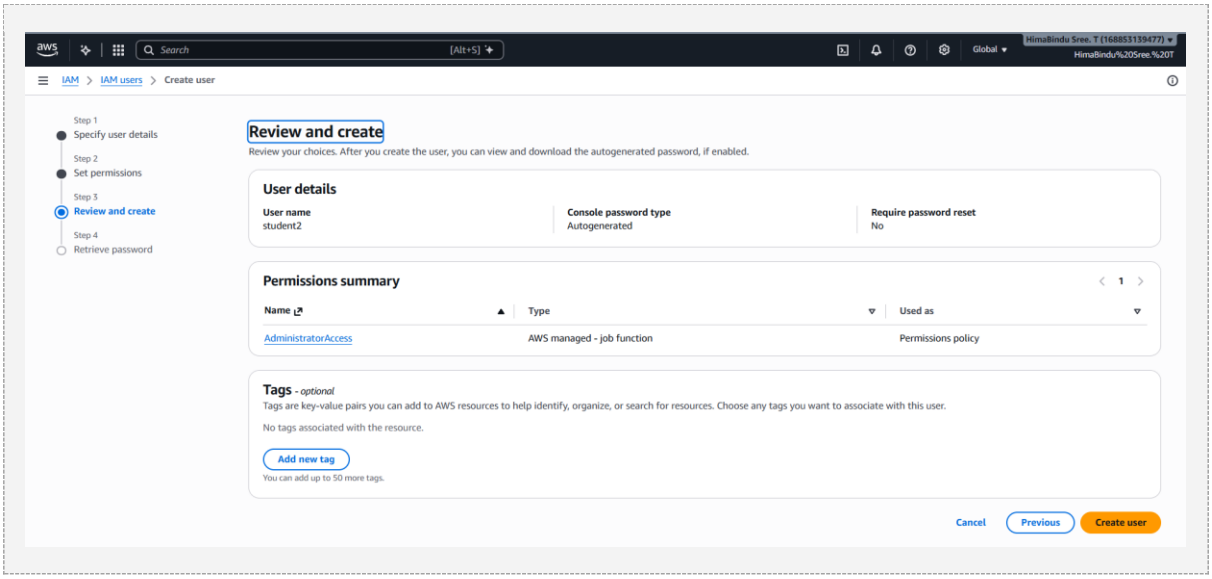
Step 4: Set the permissions

Under Set permissions I selected Attach policies directly, searched for AdministratorAccess (AWS managed – job function), ticked it and clicked Next.



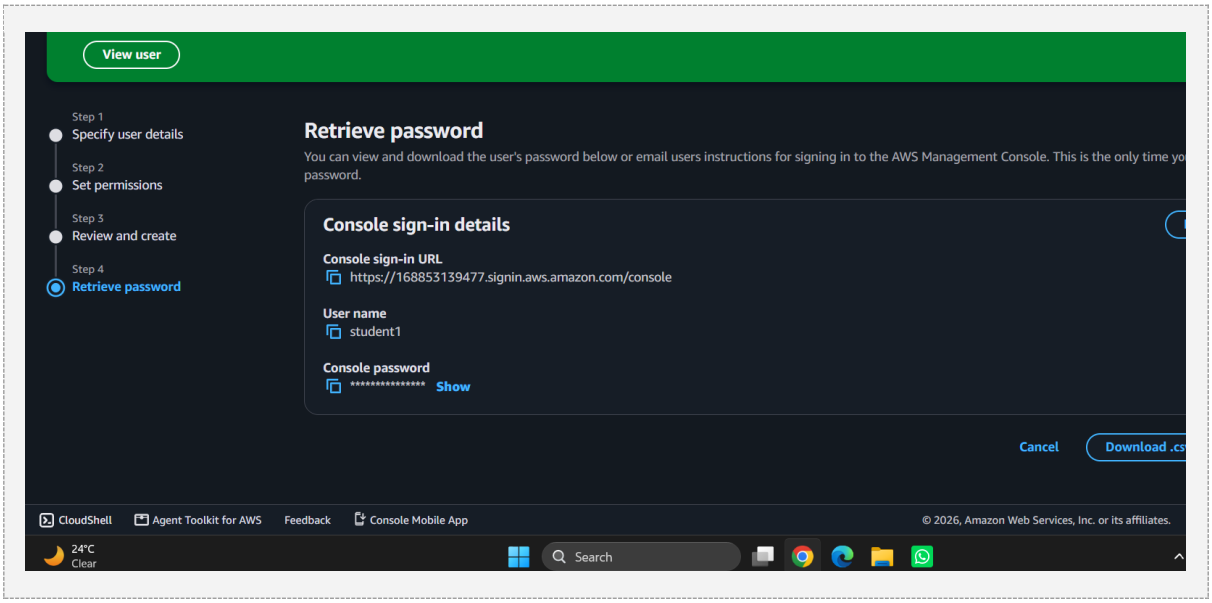
Step 5: Review and create

On the review page I checked the username and the permissions (tags are optional, so I skipped them) and clicked Create user.



Step 6: Save the sign-in details

AWS showed the console sign-in URL, the username and the autogenerated password. I downloaded the .csv file and kept it safe. Then I clicked Return to users list and saw my new user in the list.



Step 7: Sign out

I signed out of the root account.

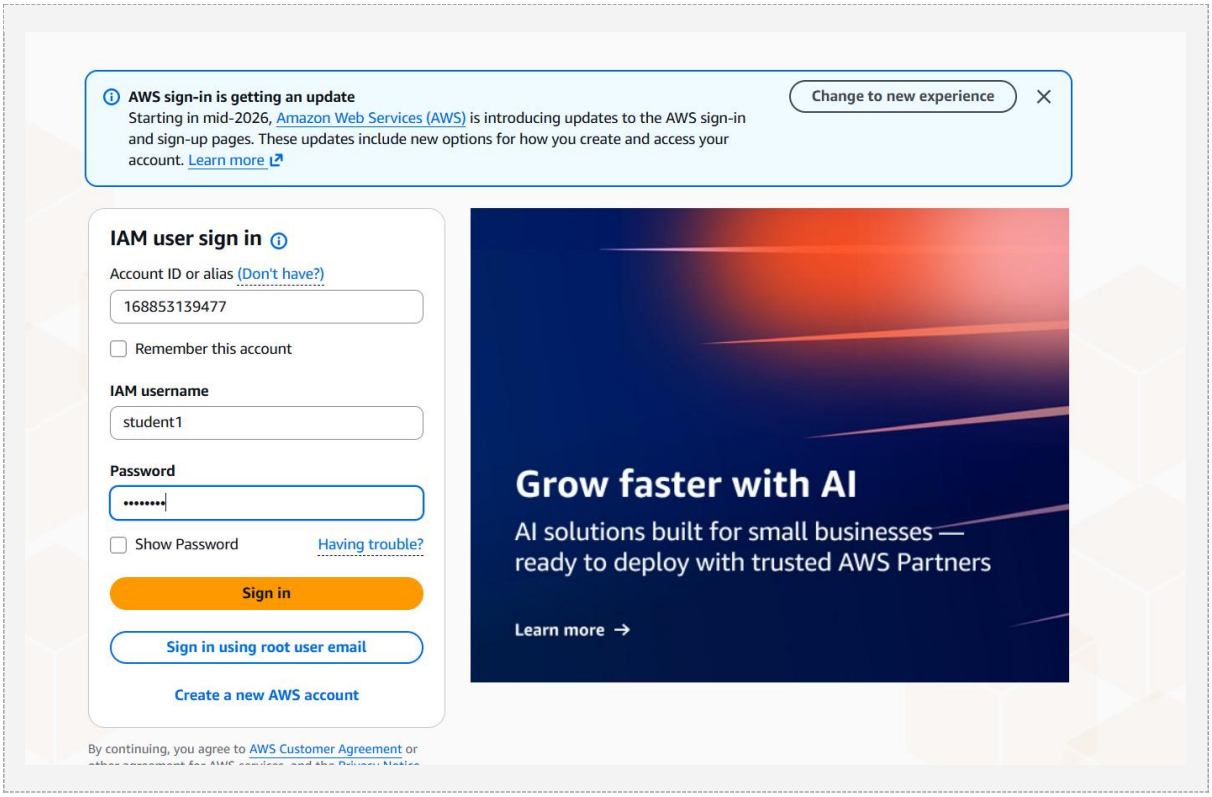
Part B: Signing in as the IAM user

Step 1: Open the sign-in page

I opened the sign-in URL from the downloaded .csv file. It opens the IAM user sign-in page with the account ID already filled in.

Step 2: Enter the credentials

I entered the IAM username and the password from the .csv file and clicked Sign in.

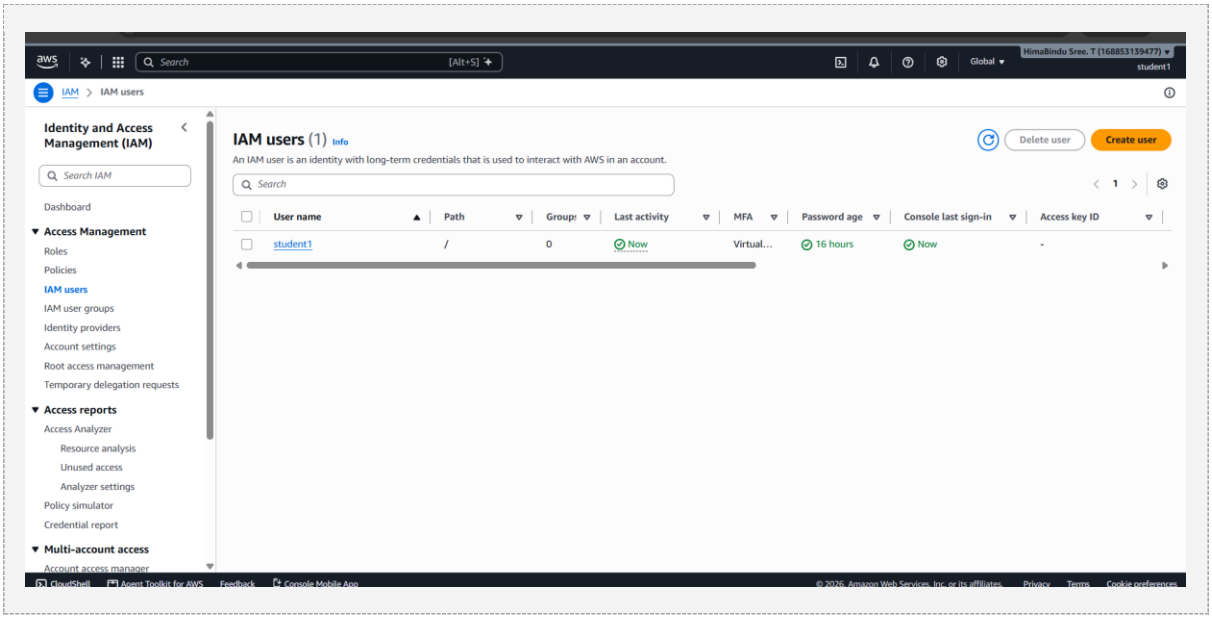


Step 3: Change the password

Because the password was temporary, AWS asked me to set a new password. I entered the old password, then the new password twice, and confirmed.

Step 4: Check the console

After this I could see the AWS Management Console, with my IAM username shown in the top-right corner. So the IAM user can access the console.



Part C: Enabling MFA

Step 1: Open the user's page

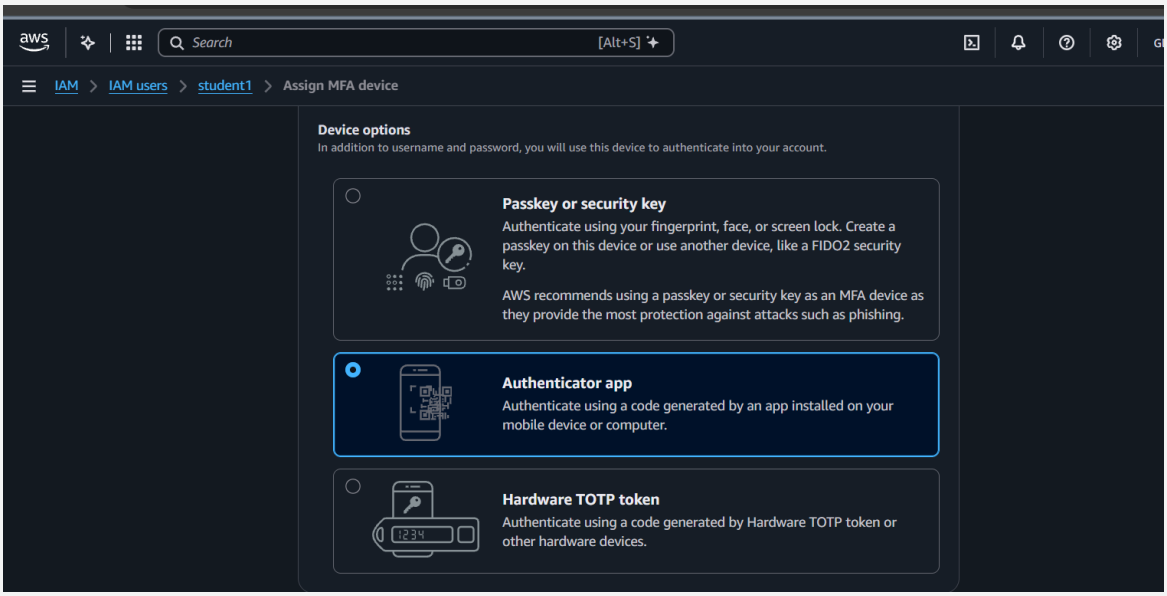
Signed in as the IAM user, I searched for IAM, opened the dashboard, clicked Users in the left menu and then clicked my IAM username.

Step 2: Go to Security credentials

I opened the Security credentials tab and found the Multi-factor authentication (MFA) section.

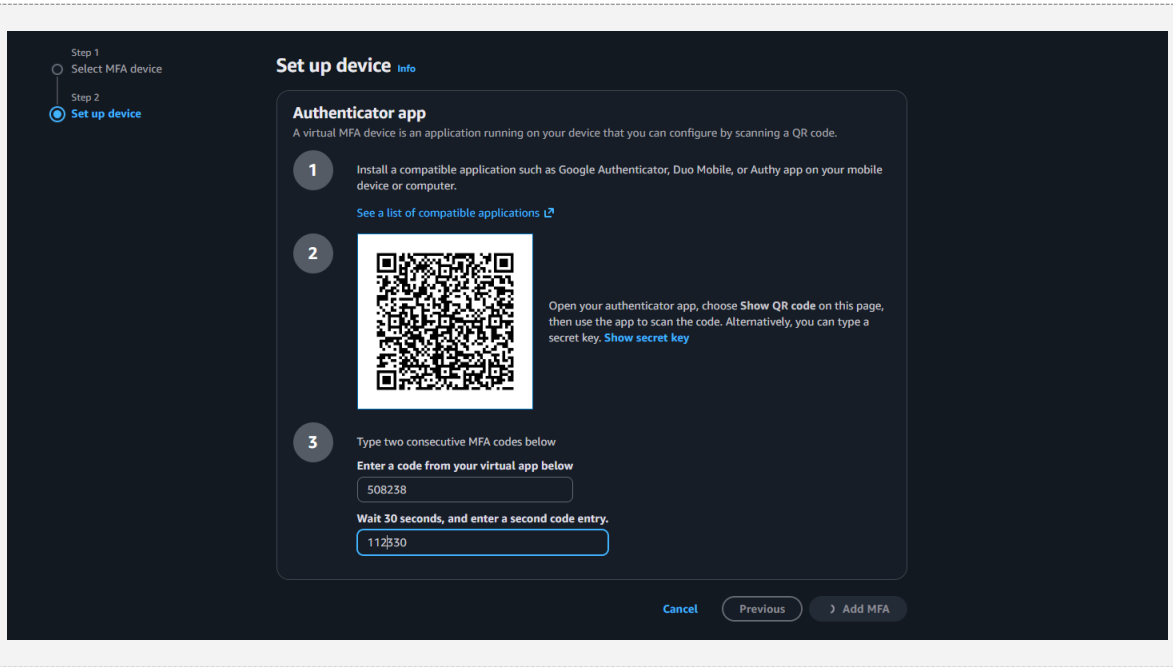
Step 3: Assign the MFA device

I clicked Assign MFA device, typed the device name (StudentMFA01), selected Authenticator app as the device type and clicked Next.



Step 4: Scan the QR code

I opened Google Authenticator on my phone, chose to add a new account and scanned the QR code shown on the AWS page.

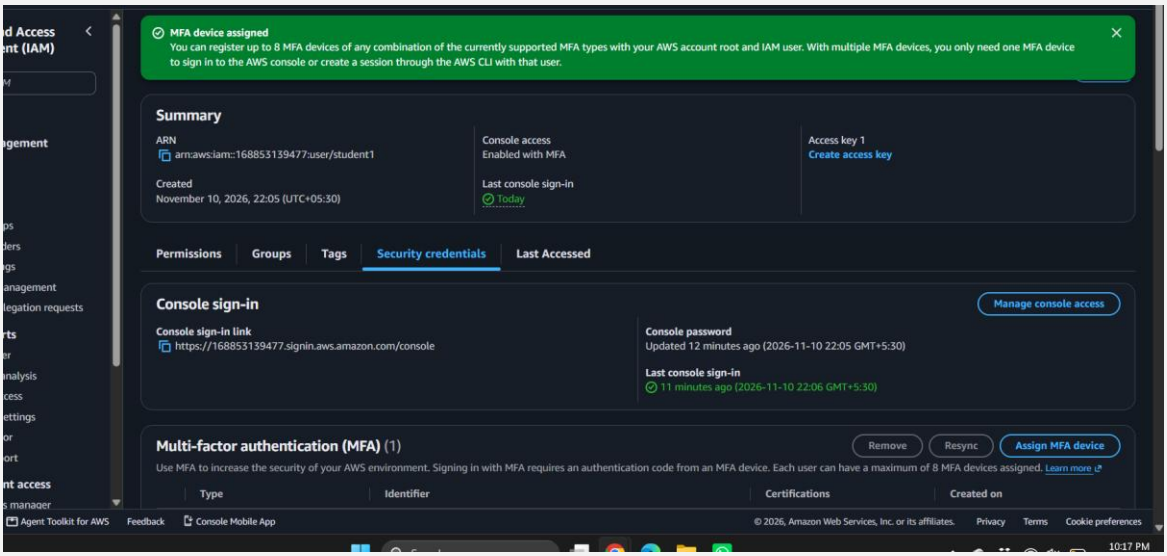


Step 5: Enter the two MFA codes

The app generated a six-digit code, which I typed in the MFA code 1 box. I waited for the app to show the next code and typed it in the MFA code 2 box. Then I clicked Add MFA.

Step 6: Verify

On the Security credentials tab the MFA device now appears as assigned to the IAM user.



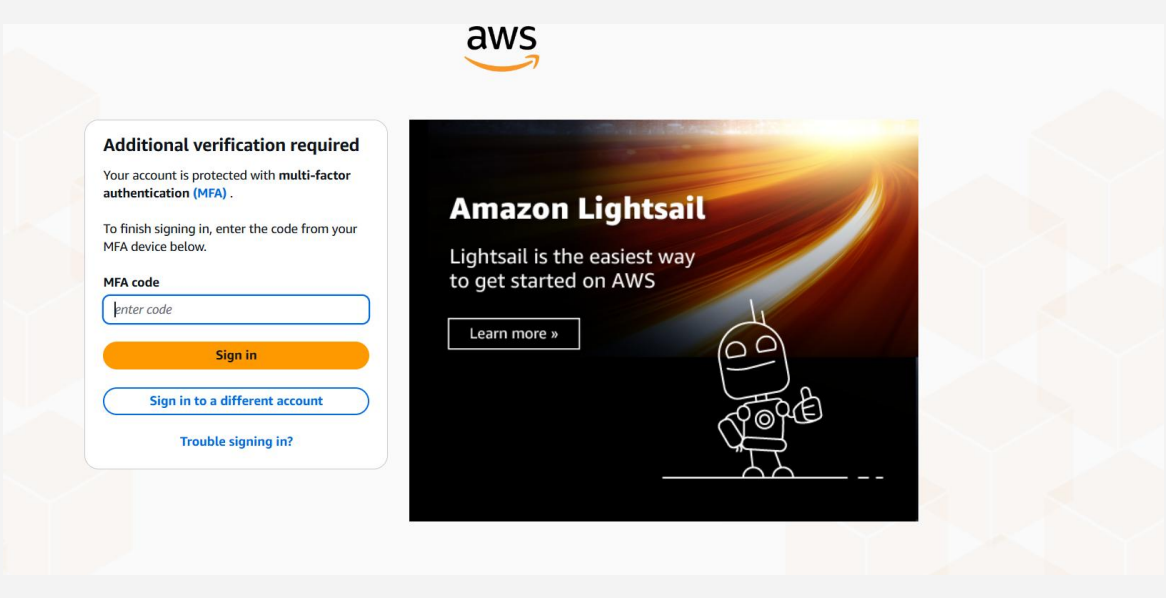
Part D: Testing MFA

Step 1: Sign out and sign in again

I signed out of the console and opened the AWS sign-in page again. I selected IAM user, entered the account ID, the IAM username and the password, and clicked Sign in.

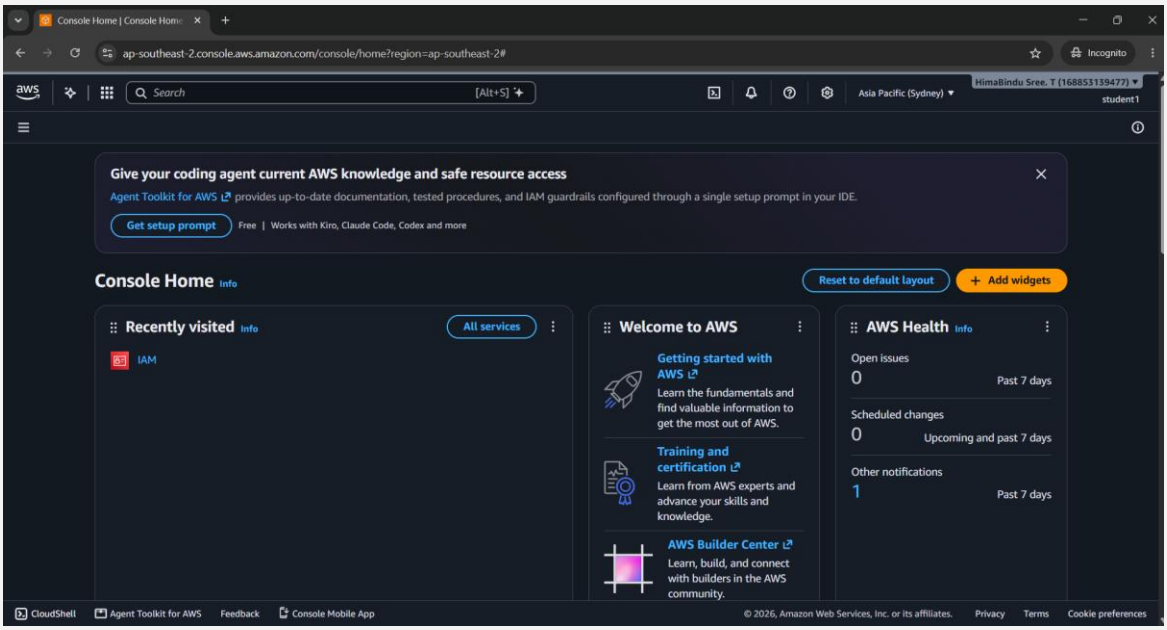
Step 2: Enter the MFA code

AWS now asked for an MFA code. I opened the authenticator app, typed the current six-digit code and clicked Submit.



Step 3: Confirm the login

After the code was accepted I was able to log in to the AWS Management Console successfully.



Result

- An IAM user was created successfully in AWS.
- The AdministratorAccess permission was given to the IAM user.
- MFA was enabled for the IAM user using an authenticator app.
- The IAM user can log in to the AWS Management Console only after entering the MFA code, which makes the account more secure.